1. General Constructor (ctor)

In C#, a constructor (often abbreviated as **ctor**) is a special method in a class or struct that initializes an object when it is created. Unlike ordinary methods, constructors do not define a return type and their purpose is to bring an object into a valid state at the moment of instantiation.

Key characteristics include:

- **Automatic Invocation**: When an instance of a type is created, its constructor is called implicitly to prepare it for use.
- **Overloading Capability**: Multiple constructors can coexist in the same type, each accepting different parameter sets, enabling flexibility in how objects are initialized.
- **Chaining**: Constructors can invoke each other using special syntax to avoid duplicating initialization logic.
- **Default vs. Explicit**: If no constructor is defined, the compiler supplies a default constructor. Defining a custom one overrides this behavior.

Constructors establish the foundation of object-oriented programming in C# by ensuring encapsulation, consistency, and reliability of object initialization.

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2. Memory Model and the Purpose of Struct in C#

C# follows a **managed memory model** governed by the Common Language Runtime (CLR). The model differentiates between two key storage regions: the **stack** and the **heap**.

- **Stack** is used for value types, function parameters, and local variables. It provides deterministic, fast allocation and deallocation due to its last-in-first-out nature.
- **Heap** is used for reference types and dynamically allocated data. Allocation here is more flexible but incurs overhead due to garbage collection and indirection.

Within this model, **structs** play a specific role. They are **value types**, which means they are stored by value and not by reference. Their main purposes are:

- **Lightweight Representation**: Structs are intended for small, immutable, and frequently used data constructs.
- **Performance Efficiency**: By avoiding heap allocation, structs reduce memory fragmentation and garbage collection load.
- **Predictable Lifetime**: Being stack-allocated (unless boxed), structs are automatically reclaimed when they go out of scope.

The existence of structs provides developers with a tool to design types that optimize memory usage and performance, particularly in performance-critical domains where overhead from reference types would be costly.

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3. Memory and Function Overloading

Function overloading in C# allows multiple methods to share the same name while differing in their parameter lists. Although this appears as a single logical entity at the source code level, at the memory and compilation level, each overloaded method is treated as a **distinct function signature**.

The implications for memory include:

- **Separate Metadata Entries**: Each overload is registered independently in the method table of the type.
- **Resolution at Compile Time**: The compiler determines which overload to call based on the provided arguments, embedding a direct reference to the corresponding method.
- **No Runtime Ambiguity**: Since resolution occurs at compile time, the runtime only needs to execute the already-bound method address, resulting in efficiency and predictability.

This model ensures that overloading does not incur extra runtime overhead, as it is essentially syntactic sugar that expands into unique method entries at the lower level.

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4. Early Binding vs. Late Binding — new vs. override

In C#, **binding** refers to the process of associating a method call with its implementation. Two forms exist:

- **Early Binding**: This occurs when the compiler determines the method to call at compile time. It provides better performance, type safety, and predictability. It applies when methods are not marked as virtual, or when a method in a derived class hides a base class method using the new keyword. In this case, the method that gets invoked depends solely on the static type known at compile time.
- **Late Binding**: This occurs when the actual method implementation is resolved at runtime, based on the object's true type. It is made possible through polymorphism, specifically when methods are declared as virtual and overridden in derived classes using the override keyword. This allows dynamic behavior and enables extensibility, since derived classes can provide specialized implementations while still being referenced through base type variables.

The choice between new and override directly impacts binding behavior:

- new indicates method hiding and enforces **early binding**, meaning the base or derived version is chosen at compile time based on reference type.
- override enables **late binding**, where the runtime determines which version to invoke depending on the object's actual type.

This distinction underscores the balance between performance and flexibility in object-oriented design: **early binding favors speed and predictability**, while **late binding favors polymorphism and extensibility**.